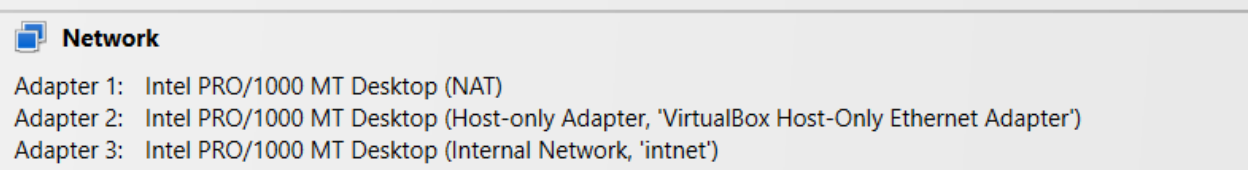
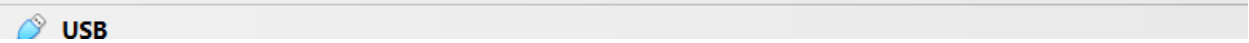
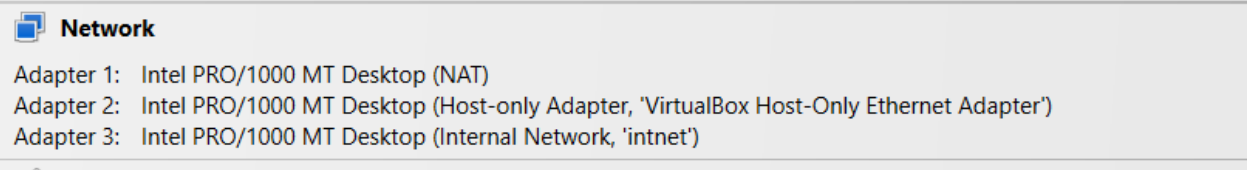


To start this lab, I made sure that my existing VM settings were set correctly to be used in later labs. Below are the Ubuntu settings followed by the windows machine settings

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Network

Adapter 1: Intel PRO/1000 MT Desktop (NAT)
 Adapter 2: Intel PRO/1000 MT Desktop (Host-only Adapter, 'VirtualBox Host-Only Ethernet Adapter')
 Adapter 3: Intel PRO/1000 MT Desktop (Internal Network, 'intnet')
- 

USB
- 

Network

Adapter 1: Intel PRO/1000 MT Desktop (NAT)
 Adapter 2: Intel PRO/1000 MT Desktop (Host-only Adapter, 'VirtualBox Host-Only Ethernet Adapter')
 Adapter 3: Intel PRO/1000 MT Desktop (Internal Network, 'intnet')
- 

USB

Next , I set up a few users and a user group in Ubuntu so that I could demonstrate hardening in the linux environment. The following was done to demonstrate hardening in the linux environment

- Created Users and Group
 - ``sudo adduser analyst1`, `sudo adduser analyst2`, and `sudo groupadd securityteam`.`
- Assigned both users to the new group with ``sudo usermod -aG securityteam analyst1`` and ``sudo usermod -aG securityteam analyst2``.
- Created and secure a shared directory:
 - ``sudo mkdir /securityteam`, `sudo chown :securityteam /securityteam`, and `sudo chmod 770 /securityteam`.`
- Inspected logs
 - ``sudo tail -n 20 /var/log/syslog`` and ``sudo tail -n 20 /var/log/auth.log``. The ``-n 20`` flag
 - `-n` was used to denote that I want to see the last 20 lines
- Searched for failed logins
 - ``sudo grep 'Failed password' /var/log/auth.log | tail -n 5``.
- Hardened the system:
 - The `/etc/ssh/sshd_config` file was edited to disable root SSH login (``PermitRootLogin no``)
- Implemented password aging
 - ``sudo chage -M 90 analyst1``
- Implemented fail2ban to defend against brute force attacks
 - `sudo apt install fail2ban -y``

I then moved on to my windows machine to demonstrate user creation, management and hardening

- Created a local user
 - Settings → Accounts → Other Users → Add someone else → 'Add without Microsoft account
- Verified the new account
 - `lusrmgr.msc` → Users.
- Configured Windows Defender
 - Windows Security → Virus & threat protection → Manage settings → enable real-time, cloud, and sample submission.
- Ran a Quick Scan and check Protection history.
- Hardened the Firewall
 - Windows Defender Firewall with Advanced Security → Inbound Rules → disable 'Remote Assistance' → create new rule to block TCP port 22 (SSH).
- I then View security events: Event Viewer → Windows Logs → Security → Filter by Event ID 4624 and 4625 to analyze logon attempts